# MIT ACADEMY OF ENGINEERING

## DSA PRACTCIAL ASSIGNMENT -07

(Trees)

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**CSE(AIML) (B1)**

TOPIC:- 1. Design a task scheduler where tasks can be added at both thefrontand rear,and high-priority tasks should be processed first.

Aim :-

To design and implement a task scheduler using a deque that allows insertion of tasks at both the front and rear and processes tasks according to their priority, with higher-priority tasks executed first.

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DESCRIPTION:- A task scheduler is designed using a Double-Ended Queue (Deque), which allows tasks to be inserted at both the front and rear. Each task is assigned a priority, and tasks are arranged in descending order of priority. The task with the highest priority is processed first and removed from the queue. This approach is useful for managing tasks efficiently in operating systems and process scheduling.

Code :

```
#include <iostream>
#include <deque>
#include <algorithm>
using namespace std;

struct Task {
   string name;
   int priority;
};

intmain() {
   deque<Task> tasks;

   //Add tasks at the rear
   tasks.push_back({"Task A", 2});
   tasks.push_back({"Task B", 1});

   //Add a task at the front
   tasks.push_front({"Task C", 3});

   //Sort tasks by priority (highest first)
   sort(tasks.begin(), tasks.end(),
      [](Task a, Task b) {
         return a.priority > b.priority;
      });

   cout<< "Task Execution Order:\n";
```

```
  while (!tasks.empty()) {
    cout << tasks.front().name
      << " - Priority: "
      << tasks.front().priority << endl;

    tasks.pop_front();
  }

  return 0;
}
```

Conclusion : The task scheduler was successfully implemented using a Double-Ended Queue (Deque) in C++. It allows tasks to be inserted at both the front and rear and processes the task with the highest priority first. This system helps manage tasks efficiently and demonstrates the use of deque operations in priority-based task scheduling.

```
--- TASK SCHEDULER ---
1. Add Task at Front
2. Add Task at Rear
3. Process Highest Priority Task
4. Display Tasks
5. Exit
Enter your choice: 3
No tasks available!

--- TASK SCHEDULER ---
1. Add Task at Front
2. Add Task at Rear
3. Process Highest Priority Task
4. Display Tasks
5. Exit
Enter your choice: 1
Enter task name (no spaces): 12
Enter task priority: 2
Task added successfully!

--- TASK SCHEDULER ---
1. Add Task at Front
2. Add Task at Rear
3. Process Highest Priority Task
4. Display Tasks
5. Exit
Enter your choice:
```